Curriculum Vitae

Basic Information

Name in full : Minho Song
Office : Science hall #125, Yonsei University, Seoul, South Korea
Research Area : algebraic combinatorics, combinatorial matrix theory

E-mail: smh3227@gmail.com

Homepage: minhosong.github.io

Positions

March 2026 -- Present	Post-doctoral fellow, Yonsei University.
September 2023 -- February 2026	Research fellow, Sungkyunkwan University.
May 2019 -- August 2023	Post-doctoral fellow, AORC, Sungkyunkwan University.
May 2018 -- April 2019	BK Post-doctoral fellow, Sungkyunkwan University.
March 2018 -- April 2018	Post-doctoral fellow, Sungkyunkwan University.

Education

March 2012 -- February 2018 Ph.D. in Mathematics, Sungkyunkwan University.
Advisor: Prof. Gi-Sang Cheon
Thesis : Riordan Lie Algebra and Extended Riordan groups

March 2010 -- February 2012 M.S. in Mathematics, Sungkyunkwan University.
Advisor: Prof. Gi-Sang Cheon
Thesis : Geometric Progression Matrices and Generalizations

March 2006 -- February 2010 B.S. in Mathematics, Sungkyunkwan University.

Research Interests

My research lies in algebraic combinatorics. I study Riordan group theory, symmetric functions, and combinatorial structures related to generalized orthogonal polynomials.

A central theme of my work is to understand how combinatorial structures reflect deeper phenomena in mathematics. In particular, I am interested in connections between algebraic combinatorics and neighboring areas such as geometry, topology, and representation theory.

I also explore probabilistic aspects and applications of combinatorics. In this direction, I have studied questions related to Kemeny's constant for finite Markov chains.

More recently, I have developed an interest in artificial intelligence for mathematics. In particular, I am interested in developing and applying FunSearch-style approaches based on AlphaEvolve as computational tools for conjecture discovery and structural exploration in mathematics.

Research Activities

Preprints

- *An explicit formula for Koornwinder moments and Rains' positivity conjecture* (with Younggwang Cho, Donghyun Kim, Jang Soo Kim, Hojoon Lee, and Jing Liu), arXiv:2606.14241.
- *Combinatorics of generalized orthogonal polynomials of type R_{II}* (with Jang Soo Kim), arXiv:2411.12345.

Published/Accepted Papers

- *Combinatorics of orthogonal polynomials on the unit circle* (with Jihyeug Jang), The Ramanujan Journal, 68:84 (2025), 1--36.
- *Combinatorial reciprocity for Riordan arrays* (with Jihyeug Jang and Louis W. Shapiro), Linear Algebra and its Applications, 721 (2025), 419--448.
- *Refined canonical stable Grothendieck polynomials and their duals, Part 2* (with Byung-Hak Hwang, Jihyeug Jang, Jang Soo Kim, and U-Keun Song), European Journal of Combinatorics, 127 (2025), 104166.
- *Kemeny's constant and enumerating Braess edges in trees* (with Jihyeug Jang, Mark Kempton, Sooyeong Kim, Adam Knudson, and Neal Madras), Linear & Multilinear Algebra, 73(8) (2025), 1529--1565.
- *Strictly monotone sequences of lower and upper bounds on Perron values and their combinatorial applications* (with Sooyeong Kim), Linear & Multilinear Algebra, 73(2) (2025), 322--350.
- *Refined canonical stable Grothendieck polynomials and their duals, Part 1* (with Byung-Hak Hwang, Jihyeug Jang, Jang Soo Kim, and U-Keun Song), Advances in Mathematics, 446 (2024), 109670.
- *A Riordan group poset* (with Louis W. Shapiro), Linear Algebra and its Applications, 689 (2024), 66--85.
- *Kemeny's constant and Wiener index on trees* (with Jihyeug Jang and Sooyeong Kim), Linear Algebra and its Applications, 674 (2023), 230--243.
- *Negative moments of orthogonal polynomials* (with Jihyeug Jang, Donghyun Kim, Jang Soo Kim, and U-Keun Song), Forum of Mathematics, Sigma 11 (2023), Paper No. e22, 34 pp.
- *Enumeration of bipartite non-crossing geometric graphs* (with Gi-Sang Cheon, Hong Joon Choi, Guillermo Esteban), Discrete Applied Mathematics, 317(2022), 86--100.
- *A new aspect of Riordan arrays via Krylov matrices* (with Gi-Sang Cheon), Linear Algebra and its Applications, 554 (2018), 329--341.
- *Finite and infinite dimensional Lie group structures on Riordan groups* (with Gi-Sang Cheon, Ana Luzón, Manuel A. Morón, Luis Felipe Prieto-Martinez), Advances in Mathematics, 319 (2017), 522--566.

Workshops Attended

- *Thematic Program on AI and Mathematics*, Korea Institute for Advanced Study, August 3--7, 2026.
- *PACOH: Artificial Intelligence for Combinatorics*, Jeju, July 6--10, 2026.
- *2025 KIAS Winter School on Mathematics and AI*, Jeongseon, December 2--5, 2025
- *Mathematical Modeling in Industry XIX*, University of Minnesota, August 5-14, 2015.

Conference Talks

- *Deograms and their links to rational Catalan objects*, ASIACOMB 2026, August 24--28, 2026, Daejeon, South Korea.
- *Moments and lattice paths for orthogonal polynomials of type R_{II}*, The 18th International Symposium on Orthogonal Polynomials, Special Functions and Applications (OPSFA-18), August 17--21, 2026, Kyoto, Japan.
- *Exact enumeration via Riordan arrays*, Soongsil University Mathematics Colloquium, April 29, 2026, Soongsil University, South Korea.
- *An invitation to Riordan group theory*, Discrete Analysis Seminar, April 7, 2026, Yonsei University, South Korea.
- *A family of pseudo-involutory functions*, Joint Mathematics Meetings 2026, January 3--7, 2026, Washington, USA.
- *A unified combinatorial framework for orthogonal polynomials including type R_{II}*, The 34th KIAS Combinatorics Workshop, May 29--31, 2025, Jeju, South Korea.
- *Two Rational Catalan Objects*, 2025 KSIAM Spring Conference, May 16--18, Konkuk University, South Korea.
- *Combinatorics of generalized orthogonal polynomials of type R_{II}*, 2025 KMS Spring Meeting, April 24--26, 2025, KAIST, South Korea.
- *Combinatorics of the orthogonal polynomials on the unit circle (OPUC)*, 2024 Combinatorics Workshop, August 28--30, 2024, Chungbuk National University, South Korea.
- *About Riordan group posets*, The 9th International Symposium on Riordan Arrays and Related Topics (9RART), June 3--5, 2024, Howard University, USA.
- *Combinatorial reciprocity for Riordan arrays*, The 30th KIAS Combinatorics Workshop, March 8--9, 2024, KIAS, South Korea.
- *Combinatorics on the negative part of Riordan matrices*, The 25th Conference of the International Linear Algebra Society, June 12--16, 2023, Madrid, Spain.
- *Enumerative results for connected bipartite non-crossing geometric graphs*, The 24th Conference of the International Linear Algebra Society, June 20--24, 2022, Ireland.

- *Enumeration of connected bipartite non-crossing geometric graphs*, 2021 KMS Spring Meeting, April 29--30, 2021, South Korea (Online).
- *Riordan group theory and connected bipartite non-crossing graphs*, 2020 Gyeonggi-Incheon Combinatorics Workshop, November 27, 2020, South Korea (Online).
- *A relation between strongly connected tournaments and connected graphs*, The 6th International Workshop on Riordan Arrays and Related Topics, July 1--5, 2019, TSIMF, Sanya, China.
- *A proof of an open problem on the commutator subgroup of the Riordan group*, Joint Mathematics Meetings, January 16--19, 2019, Baltimore, USA.
- *A new aspect of the Riordan group via Krylov matrices*, The 3rd International Symposium on Riordan Arrays and Related Topics, June 20--23, 2016, Illinois Wesleyan University, USA.
- *Lie Theory in the Riordan group*, East Asian Core Doctoral Forum on Mathematics, January 18--21, 2015, National Taiwan University, Taiwan.
- *Krylov matrices versus orthogonal polynomials*, The Second International Symposium on Riordan Arrays and Related Topics, July 14--16, 2015, Campus Lecco, Politecnico di Milano, Italy.
- *The Riordan Lie algebra*, 2014 KMS Annual Meeting, October 24--25, 2014, Yonsei University, South Korea.
- *On the commutator subgroup of the Riordan group*, The 19th Conference International Linear Algebra Society, August 6--9, 2014, Sungkyunkwan University, South Korea.
- *The Riordan Lie group and the corresponding Lie algebra*, 2014 Kangwon-Kyungki Mathematical Society Meeting, June 20, 2014, Incheon National University, South Korea.
- *Geometric progression matrices and an application to the orthogonal polynomials*, 2014 AKOOS-PNU International Conference, February 5--7, 2014, Pusan National University, South Korea.

Academic Event Organization

- Co-Organizer, *Intensive Workshop on Algebraic Combinatorics*, Department of Mathematics, Sungkyunkwan University, January 14--18, 2025.
- Co-Organizer, *Intensive Workshop on Enumerative Combinatorics*, Department of Mathematics, Sungkyunkwan University, January 24--28, 2022.

Honors and Awards

- *Sejong Science Fellowship* from National Research Foundation of Korea, March 2022 -- February 2027.
- *Creativity Challenge Research* from National Research Foundation of Korea, June 2019 -- May 2022.
- *Doctoral Dissertation Award* from Korean Mathematical Society, April 21, 2018.
- *2014 AKOOS-PNU International Conference Best Presentation Award* from Pusan National University, February 7, 2014.
- *BK21 Scholarship Program for graduate students* from National Research Foundation of Korea, March 2010 -- February 2016.
- *Global Ph.D Fellowship* from National Research Foundation of Korea, March 2010 -- February 2013.

Date: August 11, 2026